CUSTOMER PROCESS DOCUMENT

# Software Change Request Management Process

Process definition, workflow, roles, inputs, outputs, approvals and decision criteria

### Document objective

Define a clear, controlled and auditable process for submitting, assessing, approving, implementing, validating and closing software change requests. This document is intended to be understandable by business and delivery stakeholders and sufficiently structured for later conversion into a machine-consumable representation.

| | |
|---|---|
| **Document ID** | CR-PROC-001 |
| **Version** | 1.0 |
| **Status** | Draft for stakeholder review |
| **Process owner** | Change Manager |
| **Approval authority** | Change Authority / CAB |
| **Classification** | Customer Confidential |
| **Effective date** | To be confirmed |
| **Review cycle** | Annual or upon material process change |

*Illustrative customer-prepared process document. Names, roles and controls must be aligned to the customer's approved change policy before operational adoption.*

# 1. Document Control

| Version | Date | Author / Owner | Change Summary | Approval Status |
|---|---|---|---|---|
| 1.0 | To be confirmed | Change Management Office | Initial process definition | Draft |

## 1.1 Review and Approval

| Role | Review Responsibility | Approval Required |
|---|---|---|
| Change Manager | Process completeness, routing and governance | Yes |
| Application Owner | Application impact and production acceptance | Yes |
| Business Owner | Business value, priority and acceptance | Yes |
| Development Lead | Implementation feasibility and engineering controls | Yes |
| QA Lead | Test coverage and quality acceptance | Yes |
| Change Authority / CAB | Production release authorization | Yes |

# 2. Process Overview

| Attribute | Definition |
|---|---|
| Process name | Software Change Request Management |
| Business purpose | Ensure that software changes are justified, assessed, authorized, implemented, tested and closed with appropriate evidence and accountability. |
| Trigger | A stakeholder submits a formal request to modify a software product, service, configuration, interface or supporting component. |
| Start state | NEW |
| Successful end state | CLOSED |
| Alternative end state | REJECTED |
| Process owner | Change Manager |
| Primary approval gates | Business Approval and Release Approval |
| Scope | Normal software changes requiring assessment, development, testing and production authorization. |
| Out of scope | Emergency changes, pre-authorized standard changes and purely informational requests unless the customer elects to route them through this process. |

## 2.1 Guiding Principles

- Every change has a unique identifier and accountable owner.

- No change proceeds without complete inputs and defined acceptance criteria.
- Execution responsibility and approval authority are recorded separately.
- Business approval is required before development starts.
- Production deployment is permitted only after release authorization.
- All decisions, evidence and rejection reasons are retained in the change record.
- The workflow shown in this document is the authoritative source for permitted state transitions and routing.

# 3. End-to-End Workflow

**Process sequence**
NEW → SUBMITTED → ASSESSED → BUSINESS_APPROVED → TESTED → AUTHORIZED → IMPLEMENTED → CLOSED

| Sequence | Workflow Activity | Resulting State |
|---|---|---|
| START | New change need identified | NEW |
| STEP-01 | Register Change Request | SUBMITTED |
| STEP-02 | Assess Business and Technical Impact | ASSESSED |
| STEP-03 | Business Approval | BUSINESS_APPROVED or REJECTED |
| STEP-04 | Develop and Test | TESTED |
| STEP-05 | Release Approval | AUTHORIZED or REWORK |
| STEP-06 | Deploy and Validate | IMPLEMENTED or REWORK |
| STEP-07 | Close Change | CLOSED |
| END | Process complete | CLOSED / REJECTED |

*Workflow rule: a rejected business approval ends the process as REJECTED. A failed quality or release gate returns the request to Develop and Test for correction and resubmission.*

## 3.1 High-Level Decision Logic

| Decision Point | Accepted Outcome | Rejected / Failed Outcome |
|---|---|---|
| Business Approval | Proceed to Develop and Test | Record reason and move to REJECTED |
| Development and Test | Proceed to Release Approval | Correct defects and repeat STEP-04 |
| Release Approval | Proceed to Deploy and Validate | Return to STEP-04 with no-go conditions |
| Deployment Validation | Proceed to Close Change | Return to STEP-04 for correction and new release approval |
| Closure Review | Close the change | Return to STEP-06 if evidence or validation is incomplete |

# 4. Detailed Process Steps

## 4.1 STEP-01 – Register Change Request

| Field | Definition |
|---|---|
| Step type | Start / Activity |
| Owner | Requestor |
| Approval authority | Change Coordinator |
| Start state | NEW |
| End state | SUBMITTED |
| Inputs | IN-01 Change request details; IN-02 Business justification |
| Outputs | OUT-01 Registered change request |
| Standard work | Create a uniquely identified change record and capture the requested change, business reason, affected service, desired outcome, requestor and requested completion date. |
| Success route | STEP-02 Assess Business and Technical Impact |
| Failure / rejection route | STEP-99 Reject Change |
| Required evidence | Completed change request and unique CR ID |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • All mandatory fields are completed.<br>• The affected service or application is identified.<br>• The business need and requested outcome are clear.<br>• A unique change ID is generated. | • Mandatory information remains missing after clarification.<br>• The request is a duplicate.<br>• The request is outside the scope of software change management. |

## 4.2 STEP-02 – Assess Business and Technical Impact

| Field | Definition |
|---|---|
| Step type | Assessment Activity |
| Owner | Business Analyst |
| Approval authority | Application Owner |
| Start state | SUBMITTED |
| End state | ASSESSED |
| Inputs | OUT-01 Registered change request; IN-03 Current service and dependency information |
| Outputs | OUT-02 Impact assessment |
| Standard work | Assess business impact, technical impact, affected users and services, dependencies, implementation effort, security considerations, expected downtime, delivery risk and estimated schedule. |

| Field | Definition |
|---|---|
| Success route | STEP-03 Business Approval |
| Failure / rejection route | STEP-99 Reject Change |
| Required evidence | Approved impact assessment and risk record |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • Business and technical impacts are documented.<br>• Effort, dependencies and risk are recorded.<br>• Affected services and stakeholder groups are identified.<br>• The Application Owner accepts the assessment. | • The change is not feasible within agreed constraints.<br>• A critical dependency has no resolution.<br>• Risk is unacceptable and no mitigation is available.<br>• Required impact information cannot be obtained. |

## 4.3 STEP-03 – Business Approval

| Field | Definition |
|---|---|
| Step type | Approval Gate |
| Owner | Business Owner |
| Approval authority | Business Owner |
| Start state | ASSESSED |
| End state | BUSINESS_APPROVED or REJECTED |
| Inputs | OUT-02 Impact assessment |
| Outputs | OUT-03 Business approval decision |
| Standard work | Review the business justification, value, priority, funding, delivery impact and accepted risks. Record an approve or reject decision with any conditions. |
| Success route | STEP-04 Develop and Test |
| Failure / rejection route | STEP-99 Reject Change |
| Required evidence | Named and dated business approval record |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • Business value is confirmed.<br>• Scope and priority are accepted.<br>• Funding or delivery capacity is confirmed.<br>• The Business Owner records approval and any conditions. | • Business value is insufficient.<br>• Funding or capacity is unavailable.<br>• The requested scope or priority is not accepted.<br>• The change conflicts with an approved business direction. |

## 4.4 STEP-04 – Develop and Test

| Field | Definition |
|---|---|
| Step type | Engineering and Quality Activity |

| Field | Definition |
|---|---|
| Owner | Development and QA Team |
| Approval authority | QA Lead |
| Start state | BUSINESS_APPROVED |
| End state | TESTED |
| Inputs | OUT-03 Business approval decision; IN-04 Approved development baseline and test cases |
| Outputs | OUT-04 Tested software build; OUT-05 Test evidence |
| Standard work | Implement the approved change using the controlled development baseline. Complete peer review, unit testing, system testing and regression testing. Correct defects and repeat testing until the quality gate is achieved. |
| Success route | STEP-05 Release Approval |
| Failure / rejection route | Repeat STEP-04 after correction |
| Required evidence | Versioned build, review result, test report and defect status |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • Code review is complete.<br>• Required tests pass.<br>• No critical defect remains open.<br>• The build is versioned and traceable to the change ID.<br>• QA Lead accepts the test evidence. | • Required tests fail.<br>• A critical defect remains open.<br>• The delivered build does not match the approved scope.<br>• Traceability or test evidence is incomplete. |

## 4.5 STEP-05 – Release Approval

| Field | Definition |
|---|---|
| Step type | Approval Gate |
| Owner | Change Manager |
| Approval authority | Change Authority / CAB |
| Start state | TESTED |
| End state | AUTHORIZED |
| Inputs | OUT-04 Tested software build; OUT-05 Test evidence; IN-05 Release window, implementation plan and backout plan |
| Outputs | OUT-06 Release authorization |
| Standard work | Review test evidence, implementation readiness, release timing, communications, support availability and backout plan. Record a formal go or no-go decision for production deployment. |
| Success route | STEP-06 Deploy and Validate |
| Failure / rejection route | Return to STEP-04 with no-go conditions |
| Required evidence | CAB decision, approved release window and authorization record |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • Test evidence is complete and accepted.<br>• The implementation window is available.<br>• The implementation and backout plans are executable.<br>• Required support resources are confirmed.<br>• Change Authority / CAB records go-live approval. | • Evidence is incomplete.<br>• The release window conflicts with another change or blackout period.<br>• The backout plan is not viable.<br>• Required support or approval is unavailable. |

## 4.6 STEP-06 – Deploy and Validate

| Field | Definition |
|---|---|
| Step type | Implementation Activity |
| Owner | Release Manager |
| Approval authority | Application Owner |
| Start state | AUTHORIZED |
| End state | IMPLEMENTED |
| Inputs | OUT-06 Release authorization; IN-06 Approved deployment package |
| Outputs | OUT-07 Deployment and validation record |
| Standard work | Deploy the approved package within the authorized window, record implementation actions, perform technical health checks and complete agreed business validation. |
| Success route | STEP-07 Close Change |
| Failure / rejection route | Return to STEP-04 for correction and a new release approval |
| Required evidence | Deployment log, health-check result and production validation sign-off |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • The approved package is deployed.<br>• Technical health checks pass.<br>• Expected business functionality is confirmed.<br>• No unresolved critical incident is present.<br>• Application Owner signs off production validation. | • Deployment fails or only partially completes.<br>• Technical health checks fail.<br>• Business validation does not meet the agreed outcome.<br>• A critical incident or data-integrity issue is detected. |

## 4.7 STEP-07 – Close Change

| Field | Definition |
|---|---|
| Step type | End Activity |
| Owner | Change Manager |
| Approval authority | Change Manager |
| Start state | IMPLEMENTED |

| Field | Definition |
|---|---|
| End state | CLOSED |
| Inputs | OUT-07 Deployment and validation record |
| Outputs | OUT-08 Closed change record |
| Standard work | Confirm that evidence is complete, update the final change status, record lessons and follow-up actions, retain the audit trail and notify stakeholders of closure. |
| Success route | END: CLOSED |
| Failure / rejection route | Return to STEP-06 to complete validation or evidence |
| Required evidence | Closed change record, stakeholder notification and retained evidence |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • Production validation is approved.<br>• Required evidence is stored.<br>• Outstanding follow-up actions have owners and target dates.<br>• Stakeholders are informed.<br>• Closure approval is recorded. | • Production validation evidence is missing.<br>• Mandatory operational documentation has not been updated.<br>• A critical follow-up item has no owner.<br>• Stakeholder or application acceptance remains unresolved. |

## 4.8 STEP-99 – Reject Change

| Field | Definition |
|---|---|
| Step type | Alternative End Activity |
| Owner | Change Manager |
| Approval authority | Relevant Approval Authority |
| Start state | ANY ACTIVE STATE |
| End state | REJECTED |
| Inputs | Current change record; Assessment or approval evidence |
| Outputs | OUT-99 Rejected change record |
| Standard work | Record the rejection decision, reason, authority, date and any conditions for resubmission. Notify the requestor and affected stakeholders. |
| Success route | END: REJECTED |
| Failure / rejection route | Escalate to Change Manager for completion |
| Required evidence | Rejected change record and notification evidence |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|
| • The reason for rejection is explicit.<br>• The responsible approval authority is recorded.<br>• The requestor and stakeholders are notified. | • A rejection is proposed without an accountable authority.<br>• The reason is missing or unclear.<br>• Stakeholder notification is not recorded. |

| Acceptance Conditions | Rejection / Failure Reasons |
|---|---|

- The change status is set to REJECTED.

## 5. Input and Output Catalogue

| ID | Type | Information Object | Producer | Consumer / Step | Quality / Acceptance Rule |
|---|---|---|---|---|---|
| IN-01 | Input | Change request details | Requestor | STEP-01 | Mandatory request fields complete |
| IN-02 | Input | Business justification | Requestor / Business Owner | STEP-01 | Value and urgency stated |
| IN-03 | Input | Current service and dependency information | Application Owner | STEP-02 | Current and approved information provided |
| IN-04 | Input | Development baseline and test cases | Development Lead / QA Lead | STEP-04 | Controlled baseline and approved coverage |
| IN-05 | Input | Release window, implementation and backout plans | Release Manager | STEP-05 | Plans are complete and executable |
| IN-06 | Input | Approved deployment package | Development Lead | STEP-06 | Version and integrity reference present |
| OUT-01 | Output | Registered change request | STEP-01 | STEP-02 | Unique CR ID and complete request |
| OUT-02 | Output | Impact assessment | STEP-02 | STEP-03 | Impact, effort and risk accepted |
| OUT-03 | Output | Business approval decision | STEP-03 | STEP-04 | Named decision and conditions recorded |
| OUT-04 | Output | Tested software build | STEP-04 | STEP-05 | Versioned and traceable build |
| OUT-05 | Output | Test evidence | STEP-04 | STEP-05 | Required tests passed |
| OUT-06 | Output | Release authorization | STEP-05 | STEP-06 | Go-live authority and window recorded |
| OUT-07 | Output | Deployment and validation record | STEP-06 | STEP-07 | Technical and business validation complete |
| OUT-08 | Output | Closed change record | STEP-07 | Repository / Stakeholders | Final status and evidence retained |
| OUT-99 | Output | Rejected change record | STEP-99 | Repository / Stakeholders | Reason, authority and notification recorded |

## 6. Roles and Approval Authorities

| Role | Primary Responsibility | Applicable Steps | Approval Authority |
|---|---|---|---|
| Requestor | Raises the change and supplies required information | STEP-01 | No |
| Change Coordinator | Checks request completeness and coordinates intake | STEP-01 | Yes, intake completeness |
| Business Analyst | Performs impact assessment | STEP-02 | No |
| Application Owner | Accepts application impact and production validation | STEP-02; STEP-06 | Yes |
| Business Owner | Approves value, scope, priority and funding | STEP-03 | Yes |

| Role | Primary Responsibility | Applicable Steps | Approval Authority |
|---|---|---|---|
| Development and QA Team | Builds, reviews and tests the change | STEP-04 | No |
| QA Lead | Approves quality evidence | STEP-04 | Yes |
| Change Manager | Coordinates change governance, release gate and closure | STEP-05; STEP-07; STEP-99 | Yes |
| Change Authority / CAB | Authorizes production deployment | STEP-05 | Yes |
| Release Manager | Coordinates deployment and validation | STEP-06 | Operational accountability |

## 7. Business Rules and Controls

| Rule ID | Rule Name | Control Statement | Step | Failure Response |
|---|---|---|---|---|
| CR-RULE-01 | Mandatory data | A change cannot progress until mandatory request information is complete. | STEP-01 | Reject or return for clarification. |
| CR-RULE-02 | Impact acceptance | Business, technical, risk and dependency impacts must be accepted before approval. | STEP-02 | Reject when feasibility or risk cannot be resolved. |
| CR-RULE-03 | Business authorization | Development cannot start without recorded Business Owner approval. | STEP-03 | Move to REJECTED. |
| CR-RULE-04 | Quality gate | Release approval requires successful review and test evidence with no open critical defect. | STEP-04 | Correct and retest. |
| CR-RULE-05 | Human go-live authority | Production deployment requires Change Authority / CAB approval. | STEP-05 | No-go and return for remediation. |
| CR-RULE-06 | Controlled deployment | Only the authorized package and release window may be used. | STEP-06 | Stop deployment; correct and resubmit. |
| CR-RULE-07 | Closure completeness | The change closes only after validation, evidence retention and stakeholder communication. | STEP-07 | Keep open and complete missing actions. |
| CR-RULE-08 | Controlled rejection | Every rejection requires an accountable authority, explicit reason and notification. | STEP-99 | Escalate incomplete rejection record. |

## 8. State Transition and Routing Matrix

| Current State | Step | Condition | Next State | Next Step |
|---|---|---|---|---|
| NEW | STEP-01 | Request complete | SUBMITTED | STEP-02 |
| NEW | STEP-01 | Request invalid or unresolved | REJECTED | STEP-99 |
| SUBMITTED | STEP-02 | Impact accepted | ASSESSED | STEP-03 |
| SUBMITTED | STEP-02 | Impact unacceptable | REJECTED | STEP-99 |
| ASSESSED | STEP-03 | Business approved | BUSINESS_APPROVED | STEP-04 |
| ASSESSED | STEP-03 | Business rejected | REJECTED | STEP-99 |
| BUSINESS_APPROVED | STEP-04 | Quality gate passed | TESTED | STEP-05 |

| Current State | Step | Condition | Next State | Next Step |
|---|---|---|---|---|
| BUSINESS_APPROVED | STEP-04 | Quality gate failed | BUSINESS_APPROVED | STEP-04 |
| TESTED | STEP-05 | Release authorized | AUTHORIZED | STEP-06 |
| TESTED | STEP-05 | No-go | BUSINESS_APPROVED | STEP-04 |
| AUTHORIZED | STEP-06 | Deployment validated | IMPLEMENTED | STEP-07 |
| AUTHORIZED | STEP-06 | Deployment or validation failed | BUSINESS_APPROVED | STEP-04 |
| IMPLEMENTED | STEP-07 | Closure complete | CLOSED | END |
| IMPLEMENTED | STEP-07 | Closure incomplete | AUTHORIZED | STEP-06 |

**Authoritative routing rule**

The transition matrix above is the authoritative source for routing. Any future machine-consumable representation must preserve these states, conditions, approval authorities and destinations without reinterpretation.

## 9. Exceptions, Escalations and Audit Evidence

| Scenario | Required Treatment | Owner | Evidence |
|---|---|---|---|
| Missing mandatory request data | Return to requestor; reject if unresolved | Change Coordinator | Clarification log |
| Unacceptable risk or infeasible solution | Reject with accountable decision | Application Owner / Business Owner | Assessment and rejection record |
| Test failure or critical defect | Correct and repeat STEP-04 | Development Lead / QA Lead | Defect and retest evidence |
| CAB no-go | Record conditions and return to STEP-04 | Change Manager | CAB decision record |
| Deployment or validation failure | Stop, stabilize and return for correction | Release Manager | Deployment and incident evidence |
| Incomplete closure evidence | Keep change open and return to STEP-06 | Change Manager | Closure action list |

## 10. Complete-Process Acceptance Criteria

- The change request has a unique identifier and complete audit trail.
- Business and technical impacts have been assessed and accepted.
- Business approval and release authorization are recorded by the accountable authorities.
- The implemented scope matches the approved scope.
- Required testing and production validation have passed.
- No unresolved critical defect or incident prevents closure.
- Operational and stakeholder documentation has been updated as required.
- Required evidence is retained and the final status is CLOSED or REJECTED.

# 11. Machine-Consumable Conversion Readiness

This document intentionally records explicit identifiers, states, inputs, outputs, owners, approval authorities, acceptance conditions and routing destinations. These fields are intended to support later transformation into JSON, YAML, BPMN or another approved representation.

| Readiness Requirement | Status | Evidence in this Document |
|---|---|---|
| Stable process and step identifiers | Ready | Document ID and STEP IDs |
| Explicit start and end states | Ready | Sections 2, 3 and 8 |
| Inputs and outputs identified | Ready | Sections 4 and 5 |
| Owners and approval authorities explicit | Ready | Sections 4 and 6 |
| Acceptance and rejection criteria explicit | Ready | Section 4 |
| Routing for each outcome explicit | Ready | Sections 3 and 8 |
| Rules and controls separately identified | Ready | Section 7 |
| SME approval completed | Pending | Document remains Draft |
| Target machine schema approved | Pending | To be agreed before serialization |

### Conversion principle

The process should first be validated by the accountable business, application, engineering, quality and change-management stakeholders. JSON or YAML should then be generated from the approved structure. Conversion must not be treated as approval for autonomous execution.

# 12. Glossary

| Term | Meaning |
|---|---|
| CAB | Change Advisory Board or equivalent Change Authority. |
| Change Request | A formal proposal to modify software, configuration, integration, infrastructure or supporting documentation. |
| Acceptance Condition | A testable statement that must be true before a step is considered complete. |
| Approval Authority | The accountable role permitted to authorize the step outcome. |
| Backout Plan | Controlled actions to restore the last approved stable state if implementation cannot be completed safely. |
| State | The defined process status before or after a step. |
| Evidence | A record demonstrating that an activity or approval was completed correctly. |